N-C-196J
November 19, 1975
SUPERSEDING
Int. Fed. Spec. N-C-00196H(Army-GL)
June 30, 1971 and
Fed. Spec. N-C-196G
April 21, 1966

FEDERAL SPECIFICATION

CEREALS, (BREAKFAST; PREPARED, READY-TO-EAT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for ready-to-eat breakfast cereals for the use of all Federal agencies.

1.2 Classification.

1.2.1 Types, classes, and styles. Ready-to-eat cereals covered by this specification shall be of the following types, classes, and styles, as specified (see 6.1):

Type I - Wheat products.

Class 1 - Shredded whole wheat.

Style A - Large shredded wheat biscuits.
Style B - Small shredded wheat biscuits.
Style C - Loose shredded wheat.
Style D - Medium shredded wheat biscuits.

Class 2 - Pressed-flake whole-wheat biscuits.
Class 3 - Wheat flakes.
Class 4 - Malted wheat flakes.
Class 5 - Bran flakes (25 to 40 percent).
Class 6 - Bran flakes (25 to 40 percent) and raisins.
Class 7 - Wheat bran (prepared).
Class 8 - Puffed wheat (gun-puffed).
Class 9 - Puffed wheat (gun-puffed), coated.
Class 10 - Malted cereal granules.

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Type II - Corn products.

Class 1 - Corn flakes.
Class 2 - Corn flakes, coated.
Class 3 - Corn cereal (gun-puffed).
Class 4 - Corn cereal (gun-puffed), coated.
Class 5 - Shredded corn, small biscuits (oven-puffed).
Class 6 - Corn cereal (oven-puffed).
Class 7 - Corn cereal with oats, small biscuit (oven-puffed), coated.

Type III - Rice products.

Class 1 - Puffed rice (gun-puffed).
Class 2 - Rice cereal (oven-puffed).
Class 3 - Rice cereal (oven-puffed), coated.
Class 4 - Rice flakes.
Class 5 - Rice flakes or granules, protein enriched.

Style A - High rice component.
Style B - Low rice component.

Class 6 - Shredded rice, small biscuits (oven-puffed).
Class 7 - Puffed rice (gun-puffed), coated.

Type IV - Oat products.

Class 1 - Oat cereal (oven- or gun-puffed).
Class 2 - Oat cereal (oven- or gun-puffed), coated.
Class 3 - Oat cereal (other than oven- or gun-puffed).

Style A - Biscuits.
Style B - Flakes or granules.

Class 4 - Oat flakes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-1470	-	Cereals, Breakfast, Prepared (Ready-to-Eat) and Uncooked; Packaging, Packing, Labeling and Marking of.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-668	-	Sanitary Standards for Food Plants.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Human Foods; Current Good Manufacturing Practice (Sanitation).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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United States Department of Agriculture

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 20044.)

3. REQUIREMENTS

3.1 Type I, wheat products.

3.1.1 Class 1, shredded whole wheat.

3.1.1.1 Style A, large shredded wheat biscuits. Large shredded wheat biscuits shall be prepared from cleaned, sound wheat without removal of the bran. The wheat material shall be processed without flavoring agents (except that salt may be used) and shall be converted into fine shreds or filaments and formed into biscuits of uniform size and shape. The biscuits shall be properly baked to a uniform, golden-brown color. The finished biscuits shall weigh not less than 14.2 g (1/2 ounce) nor more than 28.4 g (1 ounce) each, shall be crisp, porous, and friable and shall contain no hard particles caused by consolidation of filaments before baking, except along the seam edge, if any.

3.1.1.2 Style B, small shredded wheat biscuits. Small shredded wheat biscuits shall meet in all respects the requirements specified in 3.1.1.1 for large shredded wheat biscuits, except that the wheat material shall be processed with the addition of flavoring agents and the finished biscuits shall weigh not less than 0.2 g (1/140 ounce) nor more than 0.8 g (1/35 ounce). Not more than 4 percent by weight of the cereal shall pass through a U.S. Standard No. 20 woven-wire-cloth sieve.

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3.1.1.3 Style C, loose shredded wheat. Loose shredded wheat shall be prepared from cleaned, sound wheat without removal of the bran. The wheat material shall be processed with flavoring agents, and may contain sweetening agents, converted into fine shreds or filaments of reasonably uniform length, and baked to a uniform, golden-brown color (see 6.1).

3.1.1.4 Style D, medium shredded wheat biscuits. Medium shredded wheat biscuits shall meet in all respects the requirements specified in 3.1.1.1 for large biscuits, (type I, class 1, style A), except that the finished biscuits shall weigh not less than 1.8 g (.063 or 1/16 ounce), nor more than 2.2 g (.077 or 1/13 ounce). Not more than 10 percent by weight of the cereal shall pass through a U.S. Standard No. 20 woven-wire-cloth sieve.

3.1.2 Class 2, pressed-flake whole-wheat biscuits. Pressed-flake whole-wheat biscuits shall be prepared from cleaned, sound wheat without removal of the bran. The wheat material shall be processed with or without flavoring agents, converted into layers of pressed flakes, and formed into biscuits of uniform size and shape (see 6.1). The biscuits shall be properly baked to a uniform, golden-brown color. The finished biscuits shall weigh not less than 7.1 g (1/4 ounce) nor more than 14.2 g (1/2 ounce) each, shall be crisp, porous, and friable and shall contain no hard or tough particles caused by consolidation of the pressed flakes before baking.

3.1.3 Class 3, wheat flakes. Wheat flakes shall be prepared from cleaned, sound, whole wheat. The wheat grains shall be processed with or without flavoring agents into flakes of reasonably uniform size and the flakes shall be toasted to a golden-brown color (see 6.1). Not more than 8 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.1.4 Class 4, malted wheat flakes. Malted wheat flakes shall be prepared from clean, sound, whole wheat and malted barley. The wheat grains shall be processed with the malted barley flour, with or without additional flavoring agents, converted into flakes of reasonably uniform size, and toasted to a golden-brown color (see 6.1). Not more than 8 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.1.5 Class 5, bran flakes (25 to 40 percent). Bran flakes (25 to 40 percent) shall be prepared from cleaned, sound wheat with or without partial removal of the endosperm and with or without added wheat bran. The total wheat bran component shall constitute from 25 to 40 percent of the finished product. The wheat bran material shall be processed with or without flavoring agents into flakes of reasonably uniform size, and the flakes shall be toasted to a golden-brown color (see 6.1). Not more than 20 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

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3.1.6 Class 6, bran flakes (25 to 40 percent) and raisins. Bran flakes (25 to 40 percent) and raisins shall consist of a uniform mixture of bran flakes meeting the requirements of 3.1.5, and raisins. The raisin-component shall constitute not less than 20 nor more than 40 percent of the required net weight of the product. The raisins shall be of the seedless type and shall be pliable, meaty, well-developed, of good color, and of reasonably uniform size. Raisins may be coated with vegetable oil, sugar or other materials suitable for food use to enhance the flavor, moisture retention and flow characteristics.

3.1.7 Class 7, wheat bran (prepared). Wheat bran (prepared) shall be prepared from the outer coats (bran) of cleaned, sound wheat and shall be as free from other parts of the wheat (endosperm and germ) as possible under good, modern milling practices. The bran material shall be processed into thin shreds or granules and cut or broken into pieces of reasonably uniform length not to exceed 3.8 cm (1-1/2 inches). Wheat germ and flavoring agents may be added. Not more than 15 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.1.8 Class 8, puffed wheat (gun-puffed). Puffed wheat (gun-puffed) shall be prepared from cleaned, sound, whole wheat with or without removal of the bran and without added flavoring agents. The individual grains of the finished product shall be porous and uniformly and properly expanded to approximately 15 times their original volume by the "gun-puffing" process. Not more than 2 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.1.9 Class 9, puffed wheat (gun-puffed), coated. Puffed wheat (gun-puffed), coated shall conform in all respects to the requirements for puffed wheat (gun-puffed) specified in 3.1.8, except that not more than 4.0 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve. In addition the product shall be coated with a high-boiling-point sirup of such nature that, after coating, the finished product shall not be excessively sticky. The sirup used shall be made from sugar, dextrose, corn sirup, honey or other similar sweetening agents or any combination of such sweetening agents and may contain salt. Vegetable oil may be used in processing.

3.1.10 Class 10, malted cereal granules. Malted cereal granules shall be prepared from good quality flours, including wheat flour or whole wheat flour; with malt barley flour or malt extracts, with or without flavoring agents, shortening, sugar, and milk solids (see 6.1). The flour and other ingredients shall be made into dough and baked hard to a uniform golded color. The baked product shall be processed into granules that are crunchy, of good flavor, and of reasonably uniform size. Not more than 15 percent by weight of the cereal shall pass through a U.S. Standard No. 20 woven-wire-cloth sieve.

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3.2 Type II, corn products.

3.2.1 Class 1, corn flakes. Corn flakes shall be prepared from cleaned, sound corn. The corn shall be scoured, degermed, dehulled, and processed with or without flavoring agents into flakes that are reasonably uniform in size, tender and crisp, of good flavor, and golden-brown in color (see 6.1). Not more than 8 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.2.2 Class 2, corn flakes, coated. Corn flakes, coated, shall conform in all respects to the requirements for corn flakes specified in 3.2.1 except that the product shall be coated with a high-boiling-point sirup of such nature that, after coating, the finished product shall not be excessively sticky. The sirup used shall be made from sugar, dextrose, corn sirup, honey, or other similar sweetening agents, or any combination of such sweetening agents or simple sirups and may contain salt.

3.2.3 Class 3, corn cereal (gun-puffed). Corn cereal (gun-puffed) shall be prepared from good quality farinaceous ingredients, the principal ingredient of which shall be corn. The ingredients shall be processed with or without flavoring agents, made into suitable form, and expanded by the "gun-puffing" process (see 6.1). The finished product shall be light, porous, tender, and crisp, and shall be of suitable particle size. Not more than 5 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.2.4 Class 4, corn cereal (gun-puffed), coated. Corn cereal (gun-puffed) coated shall conform in all respects to the requirements for corn cereal (gun-puffed) in 3.2.3; alternatively it shall be prepared from cleaned, sound corn, which shall be scoured, degermed and broken into particles of suitable and reasonably uniform size. The particles shall be expanded by the "gun-puffing" process to a product that is light, porous, tender, and crisp. The product shall be coated with a high-boiling-point sirup of such nature that after coating, the finished product shall not be excessively sticky. The sirup used shall be made from sugar, dextrose, corn sirup, honey, or other sweetening agents, or any combination of such sweetening agents, flavoring agents such as fruit flavors, cinnamon, cocoa, or other appropriate flavors and may contain salt (see 6.1).

3.2.5 Class 5, shredded corn, small biscuits (oven-puffed). Shredded corn, small biscuits (oven-puffed) shall be prepared from clean, sound corn that has been degermed and dehulled. The products shall be processed with flavoring agents and shall be converted into shreds or filaments and formed into biscuits of uniform size and shape. The biscuits, after oven puffing and toasting, shall be porous, tender, crisp, of a golden-brown color and shall weigh not less than 0.32 g (1/89 ounce) nor more than 0.51 g (1/56 ounce) each. Not more than 6 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

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3.2.6 Class 6, corn cereal (oven-puffed). Corn cereal (oven-puffed) shall be prepared from good quality farinaceous ingredients, the principal ingredient of which shall be corn. The ingredients shall be processed with or without flavoring agents, with or without dietary factors, made into suitable form and expanded by the "oven-puffing" process (see 6.1). The finished product shall be light, porous, tender and crisp, and shall be of suitable particle size. Not more than 5 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.2.7 Class 7, corn cereal with oats, small biscuit (oven-puffed), or other than oven-puffed) coated. Corn cereal with oats shall be prepared from high-quality corn flour and oat flour with a minimum of 33 percent corn flour and a maximum of 25 percent oat flour by weight. The farinaceous ingredients shall be made into a dough which may contain granulated sugar or brown sugar, or a combination of both, salt, and certified artificial color. The product shall be processed into strands and formed into biscuits of uniform size and shape. The product shall be coated with sirup of such nature that after coating the finished product shall not be excessively stickly. The sirup used shall be made from sugar, dextrose, corn sirup, honey, or other similar sweetening agent or any combination of such sweetening agents with vegetable oil and salt. Not more than 8 percent by weight of the cereal shall pass through U.S. Standard No. 3-1/2 woven-wire-cloth sieve.

3.3 Type III, rice products.

3.3.1 Class 1, puffed rice (gun-puffed). Puffed rice (gun-puffed) shall be prepared from cleaned, sound whole rice with the hulls removed and with or without removal of the bran. The individual grains of the finished product shall be porous, tender and crisp, and shall be uniformly and properly expanded by the "gun-puffing" process to approximately 15 times their original volume. Not more than 2 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.3.2 Class 2, rice cereal (oven-puffed). Rice cereal (oven-puffed) shall be prepared from cleaned, sound whole rice with the hulls removed and with or without removal of the bran. The product may be processed with or without flavoring agents (see 6.1). The individual particles of the finished product shall be porous, tender, and crisp, and shall be uniformly expanded by heat. Not more than 6 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.3.3 Class 3, rice cereal (oven-puffed), coated. Rice cereal (oven-puffed), coated shall conform in all respects to the requirements for rice cereal (oven-puffed) in 3.3.2 except that the product shall be coated with a high-boiling-point sirup of such nature that, after coating, the finished

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product shall not be excessively sticky. The sirup used shall be made from sugar, dextrose, corn sirup, honey, or other similar sweetening agents, or any combination of such sweetening agents, flavoring agents, such as fruit flavors or cocoa, and may contain salt (see 6.1). Vegetable oil may be used in processing.

3.3.4 Class 4, rice flakes. Rice flakes shall be prepared from cleaned, sound rice with the hulls removed and with or without removal of the bran and with or without flavoring (see 6.1). The rice grains shall be processed into flakes of reasonably uniform size and the flakes shall be toasted to a golden-brown color. Not more than 8 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve. The finished product shall be porous, tender, and crisp.

3.3.5 Class 5, rice flakes or granules, protein-enriched, styles A and B. Rice flakes or granules shall be prepared from clean, sound rice with hulls removed and with or without the removal of the bran, and with or without the addition of other farinaceous ingredients. Either casein, wheat gluten, defatted wheat germ or soyflour shall be added to raise the protein content to the proper level. The cereal shall be processed with or without acceptable sweetening and flavoring agents, salt, and vegetable oil into flakes or granules of reasonably uniform size which shall be toasted to a golden-brown color (see 6.1). The finished product shall be porous, tender, crisp. Not more than 8 percent by weight of the cereal product shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.3.5.1 Style A shall contain a minimum of 75 percent rice component on a moisture-free basis.

3.3.5.2 Style B shall contain a minimum of 55 percent rice component on a moisture-free basis.

3.3.6 Class 6, shredded rice, small biscuits (oven-puffed). Shredded rice, small biscuits (oven-puffed) shall be prepared from cleaned, sound rice with the hulls removed and with or without removal of the bran. The product shall be processed with flavoring agents and shall be converted into shreds or filaments and formed into biscuits of uniform size and shape. The biscuits, after oven-puffing and toasting shall be porous, tender, crisp, of a light golden-brown color, and shall weigh not less than 0.2 g (1/140 ounce) nor more than 0.4 g (1/70 ounce) each. Not more than 6 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.3.7 Class 7, puffed rice (gun-puffed) coated. Puffed rice (gun-puffed), coated shall conform in all respects to the requirements for puffed rice (gun-puffed) specified, in 3.3.1, except that the product shall be coated with high-boiling-point sirup of such a nature that after coating, the finished product shall not be excessively sticky. The sirup shall be made from sugar, brown sugar, dextrose, corn sirup, honey or other similar sweetening agents or any combination of such sweetening agents, flavoring agents and may also contain salt (see 6.1).

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3.4 Type IV, oat products.

3.4.1 Class 1, oat cereal (oven- or gun-puffed). Oat cereal (oven- or gun-puffed) shall be prepared from good quality farinaceous ingredients of suitable granulation, the principal ingredient of which shall be oat flour. The farinaceous ingredients shall be made into dough, with or without flavoring agents, such as fruit flavors, pressed into desired form, and expanded by the oven-puffing or "gun-puffing" processes (see 6.1). The finished product shall be light, porous, tender and crisp, and of suitable particle size. Not more than 5 percent by weight of the cereal shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.4.2 Class 2, oat cereal (oven- or gun-puffed), coated. Oat cereal (oven- or gun-puffed) coated shall conform in all respects to the requirements in 3.4.1 except that the product shall be coated with a high-boiling-point sirup of such nature that, after coating, the finished product shall not be excessively sticky. The sirup used shall be made from sugar, dextrose, corn sirup, honey, or other similar sweetening agents, or any combination of such sweetening agents, flavoring agents, such as fruit flavors or cocoa, and may contain salt (see 6.1). Vegtable oil may be used in processing.

3.4.3 Class 3, oat cereal (other than oven or gun-puffed). Styles A and B, class 3 oat cereal shall be prepared from good quality farinaceous ingredients of suitable granulation. The principal ingredient of the cereal portion shall be oat flour. The cereal shall be processed with or without acceptable sweetening and flavoring agents and dietary protein supplement (see 6.1). The finished product shall not be sticky; and after toasting, shall be porous, tender, crisp, readily friable and golden-brown in color.

3.4.3.1 Style A. Style A shall be converted into shreds or filaments and formed into biscuits of uniform size and shape. The finished product may be sweetened during processing or may be coated as applicable, by the addition of sugar or similar sweetening agents. Not more than 10 percent by weight shall pass through a U.S. Standard No. 3-1/2 woven-wire-cloth sieve. Each biscuit shall weigh not less than 0.4 g (1/70 ounce) nor more than 1.0 g (1/28 ounce)

3.4.3.2 Style B. Style B shall have not more than 10 percent by weight pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

3.4.4 Class 4, oat flakes. Oat flakes shall be prepared from good quality farinaceous ingredients, the principal ingredient of which shall be oat flour. The ingredients shall be processed with or without flavoring agents into flakes of reasonably uniform size and toasted to a golden-brown color (see 6.1). The finished product shall be porous, tender and crisp. Not more than 10 percent by weight shall pass through a U.S. Standard No. 10 woven-wire-cloth sieve.

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3.5 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 Added dietary factors. The products may contain added dietary factors, such as proteins, vitamins, and minerals, of the kinds and amounts that are acceptable under the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6.1 Additive ingredients. Any of the cereal products listed in 3.1 through 3.4 may have added to them such products as peanuts, freeze-dried fruits, marshmallows, freeze-dried ice cream and pieces of candy. When such additives are used they shall be made of high quality edible raw materials. The chemical requirements in table I shall be applicable only to the cereal portion of the product. Procurement documents shall state the kind of additive desired (see 6.1).

3.7 Chemical and physical requirements. Ready-to-eat cereals shall conform to the chemical and physical requirements shown in table I, and as specified in 3.1 to 3.6.1, inclusive.

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TABLE I. Chemical and physical requirements

Name of products	Minimum limits of Crude fiber 1/	Minimum limits of Protein (Nx6.25)	Maximum limits of Crude fiber 1/	Maximum limits of Moisture 72 hours or more after packaging
	percent	percent	percent	percent
Shredded wheat biscuits, large	1.8	-	3.0	11.0
Shredded wheat biscuits, small	1.8	-	3.0	6.0
Medium shredded wheat biscuits	1.8	-	3.0	6.0
Loose shredded wheat	1.8	-	3.0	6.0
Pressed-flake whole wheat biscuits	1.8	-	3.0	6.0
Wheat flakes	1.5	-	3.0	6.0
Malted wheat flakes	1.5	-	3.0	6.0
Bran flakes (25 to 40 percent)	3.5	-	4.7	6.0
Bran flakes (25 to 40 percent) and raisins	3.5 2/	-	4.7 2/	10.0 2/
Wheat bran (prepared)	6.0	-	-	6.0
Puffed wheat (gun-puffed)	-	-	-	6.0
Puffed wheat (gun-puffed), coated	-	-	-	6.0
Malted cereal granules	-	-	-	6.0
Corn flakes	-	-	-	6.0
Corn flakes, coated	-	-	-	6.0
Corn cereal (gun-puffed)	-	-	-	6.0
Corn cereal (gun-puffed), coated	-	-	-	6.0
Corn cereal (oven-puffed)	-	-	-	6.0
Corn cereal with oats, small biscuit (oven-puffed), coated	-	-	-	6.0
Shredded corn, small biscuits (oven-puffed)	-	-	-	6.0
Puffed rice (gun-puffed)	-	-	-	6.0
Rice cereal (oven-puffed)	-	-	-	6.0
Rice cereal (oven-puffed), coated	-	-	-	6.0
Rice flakes	-	-	-	6.0
Rice flakes or granules, protein enriched:				
High rice component	-	20	-	6.0
Low rice component	-	40	-	6.0
Shredded rice, small biscuits (oven-puffed)	-	-	-	6.0

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TABLE I. Chemical and physical requirements (cont'd)

Name of products	Minimum limits of Crude fiber 1/	Minimum limits of Protein (Nx6.25)	Maximum limits of Crude fiber 1/	Maximum limits of Moisture 72 hours or more after packaging
	percent	percent	percent	percent
Puffed rice (gun-puffed), coated	-	-	-	6.0
Oat cereal (oven- or gun-puffed)	-	-	-	6.0
Oat cereal (other than oven- or gun-puffed)	-	18.0 3/	2.0	6.5
Oat flakes	-	-	-	6.0

1/ These limiting values are on a moisture-free basis.

2/ This limiting value applies only to the bran flakes ingredient.

3/ When specified (see 3.4.3)

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3.7.1 Taste and odor. The products shall be of good characteristic taste and odor and shall be free from rancid, musty, sour, and other undesirable tastes or odors.

3.8 Workmanship. The product shall be prepared in accordance with part 128 of the Code of Federal Regulations, Title 21, Human Foods; Current Good Manufacturing practice (sanitation) in manufacture, processing, packing or holding. In addition, for Military agencies the product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection (plant sanitation inspection).

4.2.1 For civil agencies. When specified (see 6.1), inspection of plants producing the product will be conducted upon the request of the contracting officer to the appropriate Government inspection agency. Awards of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.2.2 For military agencies. The product shall be rejected if produced in plants not meeting the requirements of 3.8.

4.3 Acceptance inspection.

4.3.1 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. For applicable criteria governing inspection of containers for the Military agencies, (see 4.3.3.2).

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4.3.2 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except as otherwise stated herein.

4.3.3 Component and material inspection. In accordance with 4.1, components and material shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicalbe purchase document.

4.3.3.1 Process examination. Unless otherwise specified (see 6.1), examination shall be performed to determine conformance to specific processing and formulation requirements. Formulation records shall be maintained reflecting the ingredients being utilized in the manufacture of each of the types, classes and styles of items being procured, when such items have specific formulation requirements. In addition, type III, class 5 product formulation records shall include the quantity of rice components, and type I, classes 5 and 6 product, and the amount of bran. Nonconformance with processing or formulation requirements shall be cause for rejection of the involved finished product.

4.3.3.2 Examination of the end item. Examinations shall be in accordance with tables III and IV and 4.3.3.2.1 through 4.3.3.2.3. Sampling plans for the aforementioned tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for table IV and percent defective for table III.

TABLE II. Sampling plans 1/

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major	AQLs Minor
III	1 filled package	Packages	S-3	-	2.5
IV	1 filled package	Package	S-2	1.5	6.5

1/ Each type, class and style of cereal shall be examined as a separate lot.

TABLE III. Examination of net weight 1/

Category	Defect
Minor	
201	More than 5 percent under required net weight (containers required to have 0.45 kg (1 pound) or less)
202	More than 2 percent under required net weight (containers required to have more than 0.45 kg (1 pound)

1/ Report results to nearest 0.1 gram for individual serving containers and nearest 2.8 g (1/10 ounce) for all other containers.

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TABLE IV. Examination of product 1/

Category		Defect
Major	Minor	All types, classes and styles 2/
		Size (excluding sieve)
	201	Not size specified
	202	Weight of biscuit not within limits specified 3/
	203	Not reasonably uniform in size and shape
		Color and appearance
101		Cereal not color specified
102		Underbaked
103		Overbaked
104		Moldy
		Texture
105		Texture of cereal not as specified (e.g., not crispy, porous, tender, light, crunchy, friable - as applicable)
106		Coated cereals excessively sticky 4/
107		Biscuit has hard consolidated particles (except along seams)
		Flavor and odor
108		Not characteristic of product
109		Objectionable flavor or odor (e.g., burnt, raw, rancid, bitter, musty, sour)

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TABLE IV. Examination of product 1/ (cont'd)

Category		Defect
Major	Minor	Type I, class 6 (Bran flakes w/raisins)
110		Raisins not seedless type
111		Less than 20 percent by weight of raisins 5/
	204	More than 40 percent by weight of raisins 5/
112		Raisins tough, not meaty, or not well developed
	205	Raisins have poor color
	206	Raisins not reasonably uniform in size

1/ Presence of foreign material (e.g., filth, insects, metal, glass, wood, paint, etc.) shall be cause for rejection of the lot.

2/ See pertinent product requirement paragraph for applicability of defects. When biscuit cereals are procured, examine one per sample unit.

3/ Weighing device shall be calibrated in increments to .01 gram.

4/ To determine excessive stickiness, pour a small amount of cereal 18.3° - 23.9°C (65° - 75°F) on a piece of clean white paper and invert the paper. If any cereal sticks to the paper, it shall be considered excessively sticky.

5/ Weighing device shall be calibrated in increments to .01 gram for individual servings and 2 g (1/16 ounce) for larger size containers.

4.3.3.2.1 Examination for sieve requirements. Examination to determine compliance with sieve requirements shall be performed in accordance with 4.4. The sample for each cereal shall be a composite of approximately 50 grams derived from sufficient number of randomly selected individual servings (to the nearest whole box) to yield that quantity. For larger packages, the sample shall be one package, and the entire contents of the package shall be sieved (several sievings may be necessary). Results shall be reported to the nearest whole percent. The lot shall be rejected if result indicates non-compliance with requirement.

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4.3.3.2.2 Examination of package and labeling. Examinations shall be in accordance with the United States Standards for Condition of Food Containers.

4.3.3.2.3 Examination of waterproof case liners, packing, marking and shipping containers. Examinations shall be in accordance with the applicable quality assurance provisions of MIL-C-1470.

4.3.3.3 Testing of finished product. The finished product shall be tested to determine conformance to the pertinent characteristic listed in table I. In addition, when specified (see 6.1), a test for filth shall be performed. Test procedures shall be in accordance with 4.4. The sample for testing shall be a 227 g (8-ounce) composite of each kind of cereal derived from the number of primary packages indicated by inspection level S-2. Lot size shall be expressed in terms of the primary package. Each test result shall be reported to the same decimal place or unit as specified for the applicable requirement. The test result for filth shall be reported as pass or fail. A test failure shall be cause for rejection of the lot.

4.3.4 Final inspection. Unless otherwise specified (see 6.1), final inspection shall be made after delivery to the point of destination (see 6.4).

4.4 Examination and test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.4.1 Examination procedures.

4.4.1.1 Sieving examination. Sieving test shall be performed as follows to determine the percentage of a cereal that will pass through a U.S. Standard woven-wire-cloth sieve: Place 50 g of the cereal on a 20.0 cm (8-inch) full height U.S. Standard sieve (of the appropriate number) to which a bottom pan has been attached. Place the assembly on a smooth level surface and with a steady sieving motion move it 25 cm (10 inches) to the left and return to its original position, repeating the movement 20 times in approximately 13 seconds. The weight of the material in grams multiplied by 2 that passes through the sieve into the bottom pan is taken to be the percentage of the cereal that passes through the U.S. Standard sieve.

4.4.2 Test procedures.

4.4.2.1 Chemical analyses. Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists as indicated below:

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Test	Source	Method
Finished product:		
Moisture	Cereal Foods Section: Wheat Flour	Total solids (moisture indirect method) vacuum oven
Protein (Nx6.25)	Chapter: Cereal Foods Section: Wheat Flour	Total protein
Crude fiber	Chapter: Cereal Foods Section: Wheat Flour	Crude fiber

4.4.2.2 Filth test. All tests required to determine compliance with the provisions of the Federal Food, Drug and Cosmetic Act in regard to insects, insect fragments, rodent hairs, rodent excreta and other filth shall be made in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter - Extraneous Material: Isolation, Section - Cereal Foods, Method - Light Filth.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Civil agencies. Unless otherwise specified (see 6.1) for all levels the product shall be packaged and packed in accordance with the industry standard practice. When specified (see 6.1) the product shall be packaged and packed in accordance with MIL-C-1470 as specified for the applicable level (see 6.1).

5.1.2 Military activities. The product shall be packaged and packed in accordance with MIL-C-1470 as specified for the applicable level (see 6.1).

5.2 Labeling and marking.

5.2.1 Civil agencies.

5.2.1.1 Labeling. Any labeling that complies with Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.2.1.2 Marking for shipment. Shipping containers shall be marked in accordance with the requirements of FED-STD-123, and such other information as specified by individual agencies.

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5.2.2 Military agencies.

5.2.2.1 Labeling. Labeling in accordance with 5.2.1.1 will be acceptable.

5.2.2.2 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) The type, class, and style of product required (see 1.2.1).
(c) If the product can be furnished with or without flavoring and/or sweetening agents, whether sweetening and flavoring agents are required and the kind of flavoring agent required. (See 3.1.1.3 through 3.1.5, 3.1.10, 3.2.1, 3.2.3, 3.2.4, 3.2.6, 3.3.2 through 3.3.5, 3.3.7, 3.4.1, 3.4.2, 3.4.3, and 3.4.4).
(d) Whether dietary supplement is required (see 3.2.6, 3.4.3).
(e) The kind of additive ingredients required for a particular product (see 3.6.1).
(f) When final inspection is not to be made after delivery to the point of destination (see 4.3.4).
(g) When process inspection is not required for civil agencies (see 4.3.3.1).
(h) When filth test is required (see 4.4.2.2).
(i) When packaging and packing for civil agencies shall be in accordance with MIL-C-1470 (see 5.1.1).
(j) Selection of applicable levels of packaging and packing required (see 5.1.1 and 5.1.2).

6.2 Specification coverage. This specification does not include all types, classes, styles, and sizes of ready-to-eat cereals that are commercially available, but it is intended to cover the types, etc., that are normally suitable for Federal Government requirements and that are manufactured on a competitive basis.

6.3 If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated.

6.4 Destination inspection (for civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for identity, condition, and quantity only.

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Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
HEW-FDA, NIH
USDA-AMS
VA-DMS

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